Stability Analysis/Diffuser Design

HEV TWO

AE3-306 Group Design Project
Imperial Racing Green
Department of Aeronautics
Imperial College London

Report By: **Iman Hansra**
CID: 01192110
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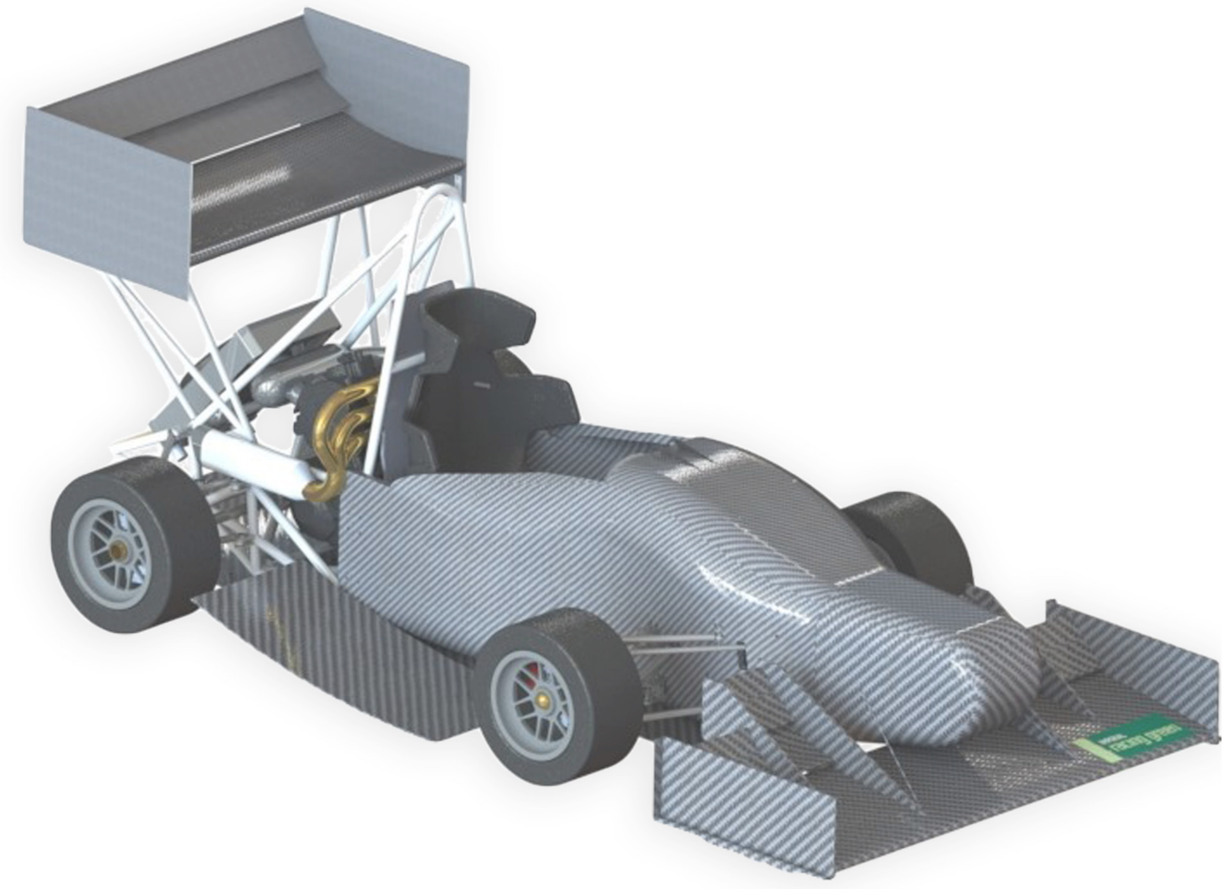
Project Coordinator: Dr Joaquim Peiro
Supervisor: Dr Joaquim Peiro



Abstract

This report focuses on two aspects involved in the design of a racing car to be entered into the Formula Student competition for Imperial Racing Green, improving upon their current design, HEV One. Firstly, the static stability of the car and its sensitivity to changes in velocity and aerodynamic centre of pressure allowed a limit to be placed on longitudinal centre of gravity and centre of pressure positions. This required the former to be as far forward as possible and the latter to be rearward of 33.5 % of the wheelbase distance. Final aerodynamic results gave a centre of pressure of 34.1 % which provides a very low stability margin. Methods are discussed for augmenting this without major changes being made to the car. Secondly, due to the absence of a diffuser on Imperial Racing Green's current design, and with this device known to have the potential to be extremely aerodynamically efficient, it was developed for the new car, HEV Two. This device, when attached to the chassis, produced downforce equivalent to an 11 point increase in the dynamic event score, accounting for its drag and mass, at a cost of just over £900. The limitations of the analysis carried out are also mentioned, including difficulties with Finite Element Analysis for the underfloor, resulting in a mass estimation being used.

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Executive summary

Introduction Imperial Racing Green will be entering their first hybrid vehicle into the Formula Student competition in July 2019. Formula Student sees global universities design and build 4-wheeled, single-seater racing cars to compete in various static and dynamic events. Their car is named HEV One and is powered by both electric motors and an internal combustion engine; it is in its preliminary design phase and has not yet been built. The task as a team of 22 is to appraise HEV One then present proposed changes as a design for a fully-formed car named HEV Two. The effectiveness of the proposal will be benchmarked against the Formula Student scoring system for which we aim to achieve the maximum possible points. It is intended that IRG use elements of HEV Two in this year's design class competition in subsequent iterations of their vehicle, and as such as a bare minimum it must comply to the FSAE Formula Student regulations whilst being manufactured and assembled within reasonable budget. The result was a 73 point increase over HEV One, scoring a total of 563 points out of 675 across all 5 events which would place HEV Two 2nd in the 2018 competition; two places higher than HEV One. *For more information, see the report written by James Bannon.*

Compliance to FS rules Each sub-group requires a thorough understanding of the Formula Student regulations which has guided certain aspects of the design. Restrictions and limitations have been put in place for the safety of the driver and to keep the competition fair and standardised. Failure to comply with any of the rules in the main or supplementary rule book can lead to the team being disqualified and unable to compete in any of the dynamic events. Regulations that affected several sub-groups needed to be considered more carefully - communication between groups was vital. *For more information, see the report written by Soe Htet Khaing.*

Vehicle Performance Modelling A laptime simulation (lapsim) tool is a non-physical model and allows quantification and comparison of performance resulting from potential design decisions. The creation and development of the lapsim was required to capture changes in physics caused by design decisions, thus allowing for design optimisation. It was essential that the model was validated in order to quantify its accuracy before its results could be trusted to guide the overall design of the HEV Two car, with the aim of maximising points over the range of dynamic events constituting the Formula Student competition. The lapsim tool was tested on twelve Formula Student cars with varying powertrain assemblies and degrees of aerodynamic package sophistication. It predicted the total points scored within 17% accuracy, which was decided to be sufficient. Therefore, various subgroups could use the lapsim to quantify design decisions, such as: evaluating the need for an aerodynamic package; vehicle layout and centre of gravity optimisation; powertrain assembly; engine configuration and suspension design to name but a few. *For more information, see the reports written by Greg Jones and Ilamaran Jayamurthy.*

Stability The stability of a race car is a key design driver since it has a significant impact on the ability of a driver to reach the limit of a car's grip potential. Stability analysis was carried out using the idealised bicycle model. The need for a positive static margin constrains the longitudinal centre of gravity position to be forward of halfway along the wheelbase, which has a significant impact on the layout of the car. Stability also limits the vertical centre of gravity location to below 400 mm via the car tilt test, whereby the car is rotated 60 degrees about the longitudinal axis and must not topple with a seated driver. Note that stability is also dependent on the distribution of aerodynamic forces and therefore, a rearward aerodynamic centre can enhance stability, although this is difficult to achieve given the placement of aerodynamic components, without sacrificing overall downforce. Dynamic stability was also carried out using equations of motion of the car for longitudinal and rotational motions during cornering. By taking an eigenvalue approach it was proved that all perturbations decay exponentially with time. *For more information, see the reports written by Iman Hansra and Nikhil Shah.*

Powertrain HEV Two has been justified to have the same baseline drive configuration as HEV One: a four-wheel drive with a through-the-road hybrid layout in which two in-hub motors power the front wheels and an independent ICE powers the rear wheels. The in-hub motors are the HEV One APS TP100 200 kv motors with a peak power of 23 kW and a peak torque of 17.6 Nm, chosen based on their ability to easily be packaged into 10" wheel rims, low cost and gaining maximum points in the motor lap simulator tests. The chosen motor controller is the BAMOCAR D3-400 200/400, changed from HEV One to correctly satisfy voltage requirements. The HV battery is of the same 12s8p layout as the HEV One battery but consists of 5 modules instead of 6, satisfying requirements while reducing weight. After evaluating alternatives, the same engine from the Triumph 675 was chosen for HEV Two. Changes to the final drive sprocket ratio resulted in a significant gain of 33 points whilst transmission modification suggestions from Imperial Racing Green were investigated and found to not be necessary. Appropriate planetary gearing for the electric motors with a ratio of 8 was chosen. A steady-state torque vectoring system using a PI controller was analysed and suggested for implementation. Interactions with other subgroups such as in-wheel packaging were vital to design. *For more information, see the reports written by Jennifer Williamson and Shakunt Tambe.*

Fuel system In HEV One, the fuel system was lifted from the Triumph Street Triple 675 motorbike. There was therefore much space for optimization by sizing the system specifically for our car. The fuel tank was resized and the material changed to fit our needs and a new fuel pump chosen to reduce volume whilst maintaining accessibility. This process led to reduced mass and volume. *For more information, see the report written by Elisa Berkane.*

Layout The aim was to successfully package the HEV Two as well as optimise its weight distribution for stability requirements, which dictated a minimum front biased distribution of 52:48 (Front:Rear), and a vertical centre of gravity as low to the ground as possible. Consideration was also given to driver ergonomics and component and subsystem integration. The solution places the key components such as the fuel tank and engine compactly behind the firewall, battery packs beneath the drivers legs and the electric motors within the wheel rims. The final design achieved a Front:Rear weight distribution of 52.7:47.3, a vertical centre of gravity 309mm above the ground and a mass saving of 32kg over the HEV One. The wheelbase and wheeltrack remain at 1550mm and 1200mm respectively, as requested by IRG.*For more information, see the reports written by Mohammed Musswer and Peter Davidson.*

Upright Design To carry the wheel hub and attach it to the suspension, an upright has been designed to do that. An upright is an essential component in all racing cars as it implements the placement of the designed suspension. Besides, the car's steering capability relies heavily on the design of the front upright as well. A thorough design optimisation, using methods like topology optimisation has been implemented to save mass and cost without compromising on its structural integrity as well as accurately accommodating for the designed suspension, preventing it to deviate from its design point. *For more information, see the report written by Ronald Teoh.*

Front chassis design The front chassis is a monocoque design, constructed from a moulded carbon fibre reinforced epoxy and Nomex honeycomb sandwich. This design was chosen as it offered the lowest overall weight against the alternatives for the required level of torsional stiffness and cost while meeting the FSUK regulations. The sandwich structure was chosen over a monolithic composite structure as it provides much more bending stiffness per unit mass, maximising the section second moment of area of the panel by separating the stiff skins with a light cellular material. The front chassis geometry was designed to balance it's torsional stiffness with overall mass and frontal area, both of which harm overall performance. Other major considerations in the chassis design were the interfaces with other systems, the manufacturability of the design geometry, compliance with FSUK regulations, driver ergonomics, systems packaging and CG location. Overall, the final design was predicted to be lighter and at least as stiff as the previous design and likely to contribute to a gain in points overall for a similar spend overall. *For more information, see the report written by Tom Findlay.*

Rear spaceframe design Rear spaceframe is one of the most fundamental components of the vehicle, as most of the powertrain and suspension systems are mounted to the rear spaceframe. By considering the cost and weight targets, steel was selected to be the rear spaceframe material rather than carbon fibre or aluminum. Several iterations were done to improve the structural performance, as well as for other components' mounting. At the end, the rear spaceframe was tested using FE simulation by applying different loading cases to make sure the frame would not fail under all conditions. The final mass of the rear spaceframe is 11.2 kg with a cost of £ 467.37. *For more information, see the report written by Jiapeng Guo.*

FEA validation and verification The Finite Element Analysis process was validation and verification, to ensure an accurate representation of real-world physics. A mesh refinement analysis was conducted to confirm that, as the number of mesh elements increased, the FE analysis was stable and converged to a finite solution. An optimal mesh size was then found; a balance between accuracy and computational resources. It was determined that a maximum mesh size of 35 mm achieved this. Validation of the FE analysis was done by replicating a 3-point bending in Creo Simulate. The simulated FE model was successful in modelling the force-displacement response in the linear-elastic region of a composite panel, with an error 0.34% when compared to experimentally obtained results. Additionally, the load at failure was also determined, deviating from experimental results by 6.02%. This suggests that the model is highly accurate and representative of real-world physics. *For more information, see the report written by Orkid Ramekaj.*

Suspension For this application, the most appropriate configuration for the suspension is the unequal length wishbones with pushrods at the front and pullrods at the rear. Liaising closely with the chassis and upright teams, the inboard and outboard suspension pickup points were calculated by assessing the kinematic response of the geometry subject to changes in bump and roll. The final suspension geometry allowed the calculation of the camber change rates which were subsequently used to determine the optimum static camber that would ensure maximum grip prior to sharp decelerations and whilst cornering. The static cambers were configured in the Lap Simulator and hence optimal camber was attained at the most critical phases of the lap. Using wheel rates optimised in the Lap Simulator, the required spring rates were calculated by consideration of the pull/pushrod motion ratios. Furthermore, the suspension geometry features anti-squat and anti-dive. The spring-damper assembly is a 4-way adjusted mono-tube with the spring setup as coil-over. Beyond the changes in camber, implementing the suspension system into the Lap Simulator proved difficult due to the simulator's 'perfect driver' assumption. Benefits of the suspension system would therefore be experienced on track with a real driver. *For more information, see the report written by Nikhil Shah.*

Computational Fluid Dynamics (CFD) validation and full vehicle simulation The CFD software used was STAR CCM+. Two validation cases were tested; the three-dimension Ahmed body, and the two-dimension Eppler E387 aerofoil. The former produced error to within 4% of reality as well as capturing the physics correctly, the latter gave up to 100% error in drag and a 3-5% error in lift. Full vehicle simulations were conducted on high performance clusters Spitfire and Hurricane. The difference between HEV One and the proposed HEV Two aerodynamic packages was an increase in downforce of 17% and decrease in drag by 14%. This constituted a projected increase of 28 points through LapSim. *For more information, see the report written by Imraj Singh.*

Front wing The front wing design was iterated from HEV One, increasing the downforce to 203 N from 149 N at 15 m/s, however it came with an extra drag penalty of 16 N, producing in total 36 N. This has resulted in the design for HEV Two with the front wing SC_L of 1.53 and SC_D of 0.27 and the overall L/D ratio of 5.6. Benchmarking against the previous iteration, the front wing has satisfied the target points gain. The design consists of a two-element airfoil configuration in front of the tires to comply with regulations and a three-element setting in front of the suspension. The wing also features an endplate, a vortex generator and a gurney flap. The chosen material was an XPRG XC110 carbon fibre composite, manufactured via the vacuum bagging process. The wing also complies with the structural deflections described in the rules. The final cost was estimated to be £1.2k with the mass of 9.32 kg, which were well below the set targets. *For more information, see the report written by Mingiyan Ushanov.*

Floor and diffuser The car upon which we are building, HEV One, did not feature an aerodynamic floor or diffuser set-up, which is the most efficient aerodynamic device on a race car, with downforce to drag ratios commonly three times larger than those of the front and rear wings. Therefore, by including one for HEV Two, significant downforce, at the cost of little drag, was added to the vehicle, giving an SC_L of 0.45, an SC_D of 0.03, and a downforce to drag ratio of 14. This increased the points in the Formula Student dynamic events by 21 in the dynamic events. The diffuser, made using a carbon-fibre epoxy reinforced polymer, was estimated to have a weight of 10 kg based on the 2018 Group Design Project, which leads to a 10 point loss in the competition, giving an overall improvement of 11 points. This will cost just over £900 to achieve, accounting for material and manufacturing costs. There were a number of interactions to consider with other components such as the drive sprocket on the rear axle and the chassis spaceframe, as well as regulatory constraints. Given these interactions, much of the diffuser design work was carried out by giving maximum freedom to other sub-groups, with the constraints lifted late on in the design process, meaning there remains room for optimisation in future iterations . *For more information, see the reports written by Iman Hansra and Drew Loynes.*

Rear wing The rear wing of the HEV Two is one of the biggest contributors to downforce on the car and is also an aid to the overall stability. The key processes in designing the rear wing involved aerofoil analysis in X-foil, geometry selection using 2D CFD and endplate design using 3D CFD. The mounting of the rear wing was chosen as a compromise between reducing the disturbance to the airflow over the suction surface of the wing and having a rigid support for the wing. The final performance values of the rear wing assembled on the full car are, 164 N of downforce and 58 N of drag. The final mass of the wing came to 6.45 kg with a cost of £526.26. *For more information, see the report written by Fahad Islam.*

Brakes Initially, complying to IRG's wishes, it was planned to reuse HEV One's brake system. However, it was found that the brake disc did not fit inside the wheel and therefore needed to be redesigned. A new, smaller brake disc was designed, allowing it to be packaged inside the wheel and to be sufficiently air cooled. This was done by changing the geometry and material to increase the surface area and therefore cooling, and increasing the maximum service temperature. The brake bias was found to be 0.68/0.32, close to the optimal given by LapSim, 0.7/0.3. *For more information, see the report written by Elisa Berkane.*

Hydraulic and Electrical system The hydraulic system is designed to transfer pedal force to the brake callipers which acts on the brake discs to stop the vehicle. As a 4-wheel braking with 2 independent circuits is required, the design of the hydraulic system revolves around the investigation of viable alternatives which draws inspiration from current vehicle hydraulic system designs to optimise HEV Two. The electrical system is in-charge of transmitting power from the battery to the areas of the car that is required. By just considering a top-level design, the systems are designed to a level which enables us to determine the components to size and place - with a final system hydraulic system mass of 5.2 kg and a final electrical system mass of 30.75 kg. Despite not using regenerative braking due to its design complexity, the potential use of it is also investigated and can potentially bring about an improvement of 15% in fuel efficiency. *For more information, see the reports written by Tyler Lee Yik Loong.*

Cooling system Five heat generating components were identified: engine, brakes, motors, motor controllers and battery. Their temperature variations were calculated for the duration of the endurance event, and it was ensured that they would not overheat, leading to a loss in efficiency or even failure of the components. Due to regulations and the rain test, it was decided to water cool the motors, as well as the engine, whilst air cooling the brakes, battery and motor controllers. *For more information, see the reports written by Elisa Berkane and Imraj Singh.*

Firewall The design of the firewall between the tractive system and the driver must comply with the regulations by being 2 layers thick with the first layer being made from Aluminium and the second layer made from a material decided using CES. The second layer also has penetration, electrically insulating and fire retardant requirements. By modelling the penetration using contact mechanics, the stresses of the plate were considered, and plate theory was used as a starting point in the design. Using the material selection knowledge and Finite Element Analysis, the plate was shown to not be able to be penetrated despite undergoing a plastic or elasto-plastic contact. By conducting design optimisation studies the final choice of material is carbon foam with a thickness of 15 mm. *For more information, see the reports written by Tyler Lee Yik Loong.*

Cost analysis The final cost estimation for HEV Two including spare parts was £21868.93 which is within the allocated budget of £25k-£30k. And the total cost excluding the spare parts is £17801.85, which is similar to past 3 years' averaged cost of £16489. To achieve the most realistic cost, the use of actual suppliers was maximised and the FSEE cost catalogue was relied on for processes' cost that was difficult to quantify. The purpose of the cost analysis team was not to only provide the cost estimation, but to guide the sub-groups financially to the selection of products and services they can work around with. *For more information, see the reports written by Jonathan Koneswaran and King Lim Keith Lee.*

Table 1: HEV Two Specification.

Wheelbase / mm	**Wheeltrack / mm**	**Mass / kg**	**L/D**	**Predicted Ranking (2018)**	**Total cost / £**
1550	1200	248	2.27	2^{nd} (563 points)	21868.93

1 Stability

1.1 Introduction

The stability of a race car is critical to its on-track performance. An unstable car will diverge away from its previous equilibrium position, and not return to it unless the car's driver is able to correct the course of the car quickly enough with steering wheel inputs. Clearly, the degree to which an unstable car can be dealt with depends on the experience of the driver, as well as the severity of the instability, but regardless of skill level, a driver who knows that they can drive to the limit of the car without having instability will be faster around a lap. Even if, for instance, a car is stable for all but one corner of a circuit, the driver will expect there to be a possibility of instability elsewhere. Therefore, it is paramount that a car possesses stability. However, too much stability results in a sluggish response and sub-optimal lap time. There is an ideal range of stability, which will be discussed later.

1.2 Background Theory

The Static Margin (SM) of a vehicle provides an indication of whether a car will experience understeer or oversteer. Understeer is defined as when a driver must apply more steering lock than expected in an attempt to make a corner's apex, whereas, oversteer requires less steering lock than expected, with the driver having to turn away from the corner to prevent a spin. Understeer is associated with stability, and oversteer with instability. In order to determine the SM of a car, it is first necessary to explain what is termed as the cornering stiffness of the tyres. This is the proportionality constant between the lateral force a tyre produces and the slip angle it experiences, with the slip angle being defined as the difference between the directions in which a tyre is pointing and travelling. After a certain slip angle is reached, the lateral force will begin to increase at a slower rate, before decreasing with slip angle - in this analysis, these effects were neglected, and it was assumed that this race car will operate in the linear region of the lateral force vs slip angle curve.

However, it is also the case that the cornering stiffness is dependent on the vertical load through the tyre. [2] provided a relation between the applied vertical load through a tyre and the corresponding cornering stiffness for the Hoosier Formula Student tyres that HEV Two will use, which fit on 10 inch wheels, but the data were at vertical loads far below those that HEV Two will experience. Therefore, data was used from the Formula SAE Tire Test Consortium [3] to obtain similar plots to those contained in [2], with the new data points being used to fit a curve, using [4], that can be applied for any vertical load through tyres. Figure 1 displays the raw data used, with the gradients of the linear regions of the curves representing the cornering stiffness, a negative quantity. The fitted curve is represented in Figure 2. Note that the Bicycle Model, used for this stability analysis, by definition assumes a car with two wheels - the front wheels are joined together into one single wheel, with the same for the rear of the car (see Milliken [5] for more detail on the bicycle model). However, when using Figure 2 to find the cornering stiffness, it was assumed that each tyre at the front takes half of the vertical load on the front axle, before this was doubled to find a total front cornering stiffness that is compatible with the bicycle model. The same method was applied for the rear of the car.

Returning to the Static Margin, its value can be defined by considering the Neutral Steer Point (NSP), which is the longitudinal position of the centre of gravity between the front and rear wheels at which an applied lateral force (e.g. due to acceleration during cornering), will produce no rotational velocity about the car's centre of gravity. Intuitively, one may consider this point to be halfway between the wheels, but this assumes a neutrally stable car and no aerodynamic loading. The NSP can be defined using Equation 1, where C_R and C are the rear and total cornering stiffness respectively, the latter being the sum of the front and rear cornering stiffness.

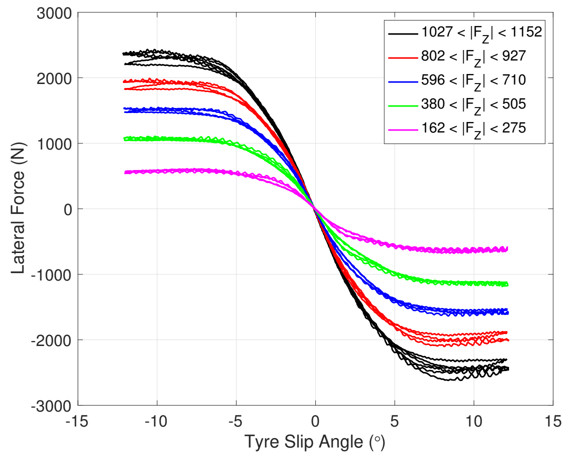

Figure 1: Data used to find Tyre Cornering Stiffness

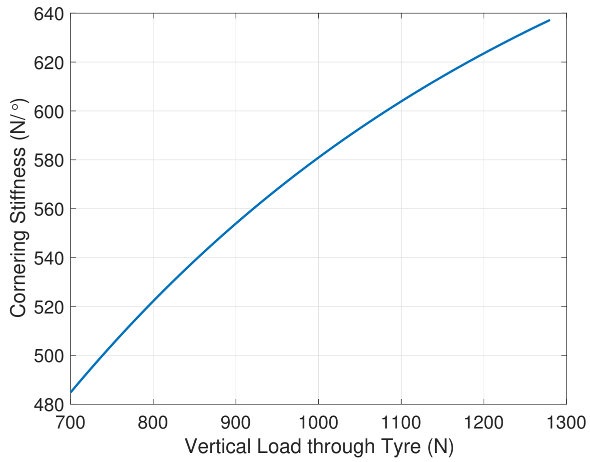

Figure 2: Cornering Stiffness vs Vertical Load relationship

$$NSP = \frac{C_R}{C} \quad [5] \tag{1}$$

From Equation 1, due to the NSP dependency on tyre cornering stiffness, there is also a dependency on the vertical load through the tyres. Therefore, if the centre of gravity is not halfway between the wheels and aerodynamic effects are considered, the vertical load distribution will not be equal front-to-rear, creating a difference in the front and rear tyre cornering stiffness. This moves the NSP away from 50% of the distance between the front and rear axles. The Static Margin is defined relative to this position - if it is forward of the NSP, then stability results. The following equation is used to determine SM, where a represents the distance from the front axle to the centre of gravity and l is the car's wheelbase.

$$SM = NSP - \frac{a}{l} \quad [5] \tag{2}$$

It has already been mentioned that excessive stability is almost as undesirable as instability, with [6] suggesting a static margin of 3-7% for a vehicle, although for a race car it can possibly be lower than this given that a comfortable driving experience is not a priority. Clearly the desired value will depend on the quality of the driver, for instance, and is also complicated by the layout of a Formula Student car, which features a rearward driver and engine position, moving the longitudinal centre of gravity position further rearward. There had been no stability analysis conducted for HEV One, and this was evident from the fact that initial estimates for this car's longitudinal centre of gravity position placed it behind halfway between the wheels, thereby producing an unstable car. However, Davidson [7] ensured that the centre of gravity moved forward to accommodate stability, and he explains in detail how this was achieved.

1.3 Sensitivity Analysis

aerodynamic sensitivity analysis was conducted using the target downforce coefficient of 3, with the effects on the SM of both the velocity of the car, and hence the magnitude of the aerodynamic force, as well as the position of the aerodynamic force longitudinally being considered. The results of this analysis are displayed in Figures 3 and 4. An aerodynamic centre of pressure of 40% of the wheelbase was predicted before development began based on the longitudinal positions of the car's main downforce-producing components: the front and rear wings, as well as the underfloor.

Note that Davidson moved the centre of gravity as far forward as possible (52.7 weight on the front axle was achieved) because, as can been seen from Figure 4, with an aerodynamic centre of pressure of 40% of the distance between the wheels, the car would only have a minor stability margin to allow for any variation in the centre of pressure during bodywork development. Note that absolute downforce was prioritised over the centre of pressure position (from a stability point of view) given the sensitivity to the former was much greater than to the latter, determined by Jayamurthy [8] from the Lap Simulator.

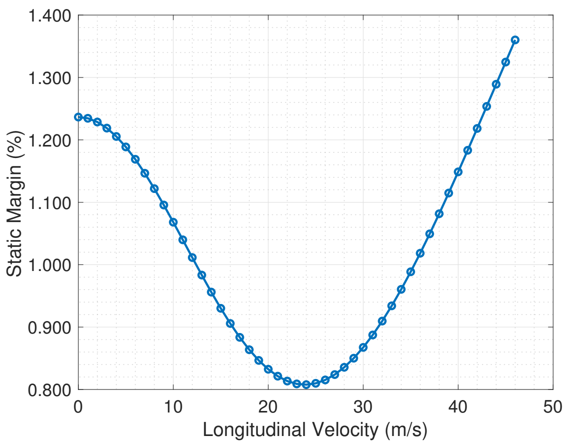

Figure 3: Effect of Velocity on Static Margin for Centre of Pressure of 40% of Wheelbase

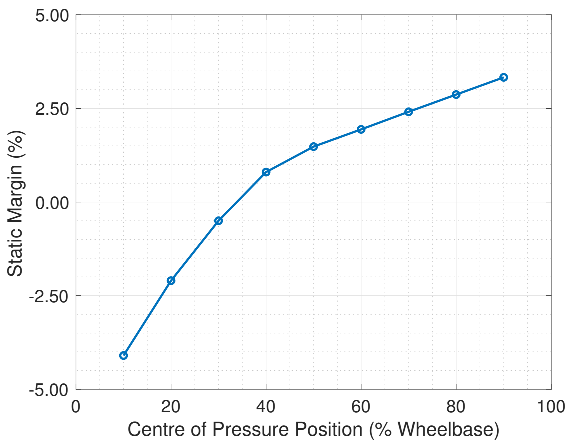

Figure 4: Effect of Centre of Pressure on Minimum Static Margin

1.4 Final Comments on Stability and Future Work

The results of the full car aerodynamic simulations produced a centre of pressure of 34.1% which, as seen from Figure 4, only provides a very small stability margin, and perhaps an unsatisfactorily small one. There are three ways in which this problem could be tackled. The first is to add weight to the front of the car to increase the stability margin, but this has a significant performance penalty from the Lap Simulator, with 1 kg extra causing a 1 point loss in the Formula Student dynamic events. Secondly, it would be feasible to make the front wing flap adjustable such that, if the driver feels some instability while racing, a reduction in front wing flap angle could be requested, resulting in a more rearward aerodynamic balance, and a stable car. Thirdly, the cornering stiffness of the rear tyres could be increased, moving the NSP rearward, by using either a different specification of tyre on a 10 inch wheel, or a larger wheel size. However, given that Imperial Racing Green have already purchased the tyres, this is not a feasible option, making front wing adjustability most appropriate. Note that the MATLAB code used for stability analysis has been added to the Lap Simulator as an output so that any changes to car parameters in future will produce a direct output of whether the car is stable.

In future, this stability model could be enhanced by incorporating the longitudinal and lateral load transfer models that were developed for the Lap Simulator [9], allowing further degrees of freedom, namely the pitch and roll angles, to be accounted for. In addition, with greater aerodynamic testing at different conditions that the car will experience during a corner, such as a yawing or nose down condition, a model for the downforce coefficient could be produced to again improve the accuracy of the stability outputs. Furthermore, during track running, using a steering wheel angle plot through the course of a lap (via car telemetry), and subsequently finding the angle of the front wheels via the steering ratio, it would be possible to compare this graph to that obtained using a neutral steer car (neither understeer nor oversteer) via the Equation , where R is the radius of each corner around the circuit in question.

$$\delta_{neutral} = \frac{l}{R} \quad [10] \tag{3}$$

If the steering angle is greater in magnitude than that found from Equation 3, then the driver is experiencing understeer, and if the angle is lower, then oversteer is present [10]. The difference between the actual and neutral steer plots can then be used to quantify what the driver is feeling, especially if they are inexperienced.

2 Diffuser Design

2.1 Introduction

A floor-diffuser aerodynamic device is considered to be the most efficient on a race car in the sense that it has the potential to produce more downforce for a given amount of drag than either a front or rear wing. The HEV One Formula Student car did not feature such a device, so with its efficiency in mind, this was a key focus of development in the aerodynamic sub-group. Part of the efficiency of the device arises from its use of ground effect, but this also causes an additional degree of freedom to be introduced into the optimisation problem that is not, for instance, present for the rear wing.

2.2 Constraints

There were a number of constraints to consider in the design of the floor-diffuser, both from a regulatory point of view and also from considering the interactions with other sub-groups. The regulatory limits apply in a number of ways, such as for the height, width and length of bodywork, and exclusion zones around the wheels, as well as a minimum ground clearance of 30 mm with the car stationary. Loynes [11] provides more information on this. Other components, such as the spaceframe and the powertrain were significant factors in determining the available expansion room for the diffuser, and after 2D analysis, the worst case scenario for the diffuser expansion was taken, whereby the sprocket producing drive from the engine to the rear axle was to be maximum size and the lower spaceframe element was to be placed underneath/behind this sprocket, as shown in Figure 5. While this restricts design potential, it is also supported by the Lap Simulator, in that increasing the transmission ratio of the engine (achieved by having a larger sprocket on the rear axle) is more of a performance improvement than could realistically be achieved by having a diffuser expansion slightly further forward. More detail on the transmission ratio can be found from Tambe [12]. The diffuser analysis which follows assumes that the expansion angle is constant across its span, which need not be the case, but was done in order to constrain a degree of freedom in the initial design process. Loynes [11] tested a diffuser with more aggressive outer channels but this was not a positive development. This is something that could be explored in more detail in future iterations of this design, particularly given that competitor teams utilise similar geometry. Ultimately, as is discussed later, the final constraints were not quite as severe as this, which enabled a small performance improvement to be realised.

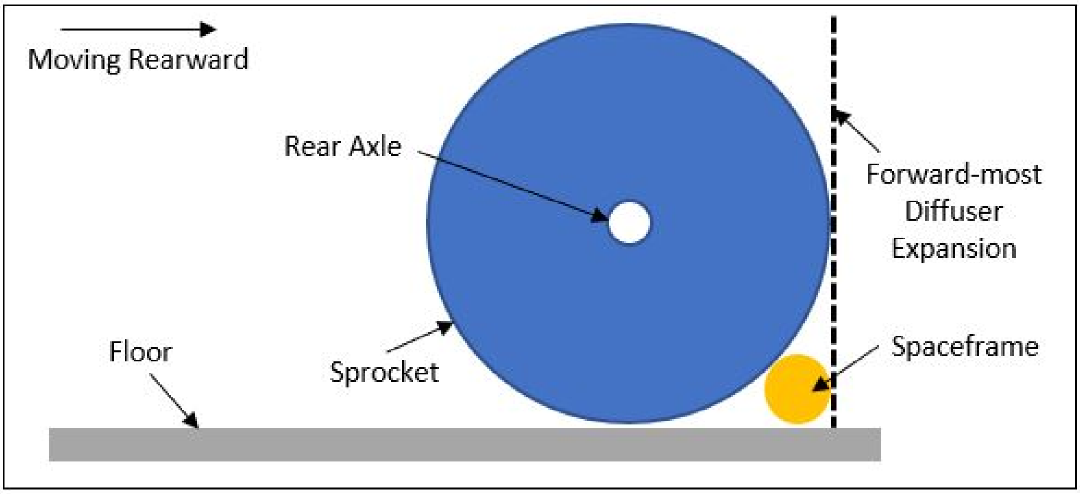

Figure 5: Interactions between diffuser, drive sprocket and spaceframe

2.3 Two Dimensional Diffuser Analysis

It was decided that the diffuser would begin expanding 160 mm behind the rear axle due to the constraints outlined above, leaving a total expansion length of 290 mm. The expansion was chosen to be gradual, with a constant radius round that just reached the trailing edge of the diffuser while maintaining the 290 mm expansion, based on the success of the 2018 Group Design Project diffuser design [13]. Two dimensional (2D) analysis was used to determine whether a lower ground clearance, coupled with a lower expansion angle, or a higher angle of expansion together with a higher clearance would yield more downforce without significant separation. Note that 2D Computational Fluid Dynamics (CFD) was validated by Singh [14], giving particularly good results for the downforce coefficient, which is the primary basis for the diffuser analysis which follows. Drag is of secondary importance given that an aerodynamic device using ground clearance is already very efficient, and that the sensitivity to a unit increase in drag is much lower (by a factor of two) than the sensitivity to a unit increase in downforce (see Jayamurthy for more detail [8]). This means that so long as a development has a downforce-to-drag ratio of greater than 0.5, it will produce an increase in performance, not accounting for weight. However, given that the diffuser downforce-to-drag ratio is often greater than 5 and the mass was low (5 kg) based on the 2018 Group Design Project diffuser reports [13] [15], it was determined that drag did not need to be considered.

Furthermore, only the underside of the diffuser was accounted for at this stage due to the fact that the top surface would feature multiple obstacles such as the chassis, engine and suspension members, while the lower surface is more critical from a flow quality perspective. In this way, a fluid volume was defined that used an inlet flow to the underside of the floor of 15 m/s and considered a 2D duct flow bounded by the underside of the diffuser from above and the moving ground from below. A 15 m/s test velocity was chosen in order to permit a consistent benchamrk with Imperial Racing Green's analysis of HEV One. A number of different expansion angles were tested using the minimum ground clearance before significant separation was present. The downforce coefficient increased with a lower ground clearance even with some separation, but eventually, it was decided to prioritise flow quality over purely looking at downforce coefficient numbers, and when the reversed flow velocity in the longitudinal direction exceeded 0.2 m/s, the minimum ground clearance was deemed to have been reached. This reversed flow velocity also coincided with a noticeable recirculating eddy appearing in the velocity vector cross-section. The results of the 2D simulations are displayed in Figure 6.

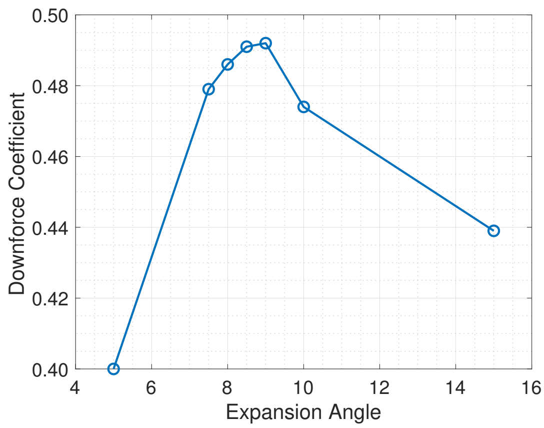

Figure 6: Results of 2D diffuser analysis

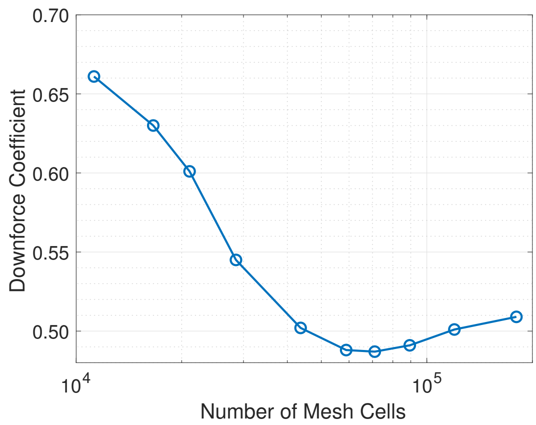

Figure 7: 2D CFD Mesh Convergence

For very low angles of expansion, the downforce coefficient is low because the optimum ground clearance before no separation is achieved is at lower than 30 mm, which is not compatible with the regulations. At the same time, higher angles of expansion with a higher ground clearance before significant separation are also not optimal. The optimal case is when the ground clearance is minimised (30 mm) and the angle is chosen such that separation is not significant (9 degrees in this case). This set of results was then used for the initial 3D analysis, removing a degree of freedom from the design process. Finally, note that CFD mesh convergence was achieved, as explained by Singh [14] and shown in the 2018 Group Design Project diffuser report by Patel [13], with Figure 7 providing evidence of this. The 2D analysis was carried out with around 90,000 cells, more than was required in hindsight, given that this mesh convergence analysis was carried out after the analysis was run. Therefore, this process could be optimised if run again.

2.4 Three Dimensional Diffuser Analysis

2.4.1 Expansion Angle

Three dimensional (3D) analysis involved finding the maximum angle of expansion before significant separation was present, again using a downforce coefficient and flow quality approach to achieve this, at the desired ground clearance of 30 mm. In 3D, it is less intuitive to define a reversed flow velocity at which separation is called significant, as was the case for 2D, as the 3D nature of downforce-producing bodies means that there is a spanwise variation in the flow field. Specifically, due to the vortices produced at the edge of the diffuser, there is more upwash as one moves outboard on the underside of the diffuser, meaning that the centre of the diffuser is most likely to separate first, something which was backed up using CFD wall shear stress plots. These showed a negative wall shear stress in the centre of the diffuser near the trailing edge, but a positive value further outboard. The results of the variation in the diffuser angle are displayed in Figure 8.

From Figure 8, it is evident that the downforce coefficient increases sharply with expansion angle until the angle reaches 13 degrees, at which point the benefit of a higher angle is minimal, with no improvement after 15 degrees. This is similar in shape to an aerofoil downforce coefficient versus angle of attack curve, where small separation can still yield an improvement in downforce before the aerofoil eventually stalls and its downforce-producing capacity drops sharply. Therefore, it was decided to use a 13 °expansion angle, with this providing a small safety factor due to the approximate nature of CFD simulations and, at this stage, a lack of consideration of upstream

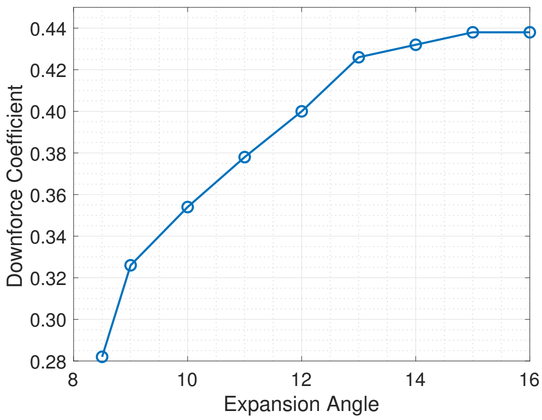

Figure 8: Results of 3D diffuser analysis

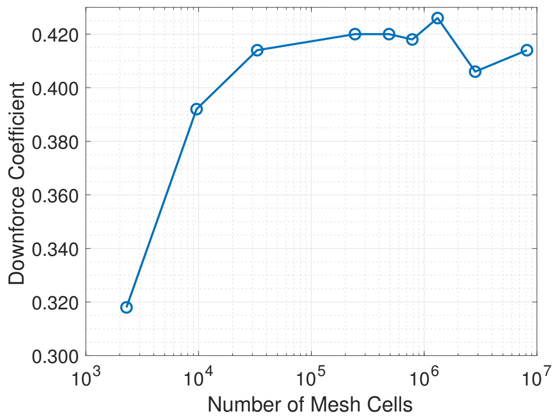

Figure 9: 3D CFD Mesh Convergence

aerodynamic behaviour from devices such as the front wing and nose. Figure 9 confirms mesh convergence for the 3D diffuser analysis in a similar manner to that described for 2D. Around 1.3 million cells were used for this analysis, again proving to be more than required to achieve mesh convergence, meaning that this process, if used again, could also be streamlined.

2.4.2 Impact of Tyre Aerodynamics on Diffuser Performance

With what could be considered as the primary parameters of the diffuser having been determined given the constraints placed upon the design, the impact of the tyres on the aerodynamic performance of the floor and diffuser was considered. Their inclusion was expected to worsen the results in terms of the downforce coefficient produced, and this was very much the case. Considering the front tyres, they produce a wake, part of which is sucked under the floor, lowering the energy (equivalent to total pressure) of the air under the floor and diffuser (see Figure 10), thereby reducing its potential to create downforce. The rear tyres were seen to have a more destructive effect, given that, unlike the front tyres, the effect they have on the upstream flow impacts the diffuser performance. Like the front tyres, there is a static pressure build up on the front face of the tyre, which is effectively a bluff body in the free stream, but one that also rotates. This pressure increase is particularly prevalent just ahead of the contact path of the tyre with the ground, as air is also being rotated into the ground by the tyre's movement. This high static pressure was then seen to cause a corresponding rise in the static pressure under the end of the floor and start of the diffuser, unloading the device despite there also being an increase in static pressure above the floor. Figure 11 provides a static pressure plot of the underside of the diffuser to display this effect.

This increase in static pressure also creates a streamwise adverse pressure gradient, slowing flow under the floor and reducing the effectiveness of the diffuser. At the same time, the high static pressure on the rear tyre produces information to the flow upstream that it will need to curve around either side of the tyre, and this process causes acceleration of the flow before it eventually detaches. As a result, there is a drop in static pressure both above and below the diffuser, but the effect is more dominant on the top surface, again reducing downforce. This second effect was less prevalent for the front tyres due to the narrower floor inboard of the tyre to allow for the steering angle of the tyres, which Loynes [11] details. Furthermore, due to the proximity of the rear tyre to the diffuser, the vortex produced along the floor's edge is much less dominant, so as the vortex core rolls up on to the underside of the diffuser, its suction is less powerful. A summary of the impact of the tyres is displayed in Table 2, with the rear tyre clearly having the most impact - note that due to the tyre affecting both the flow above and below the diffuser, these downforce coefficients include the top surface.

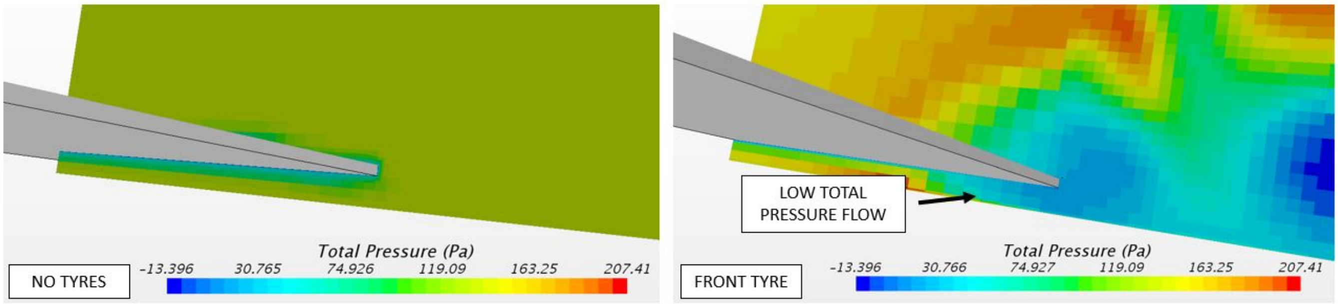

Figure 10: Total pressure plane 1 m behind front axle relative to ambient static pressure

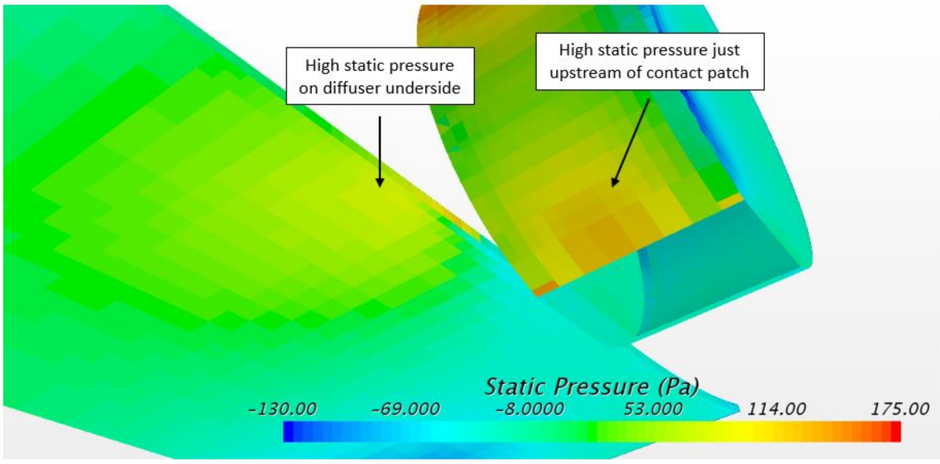

Figure 11: Static pressure surface plot of diffuser and rear tyre relative to ambient pressure

Table 2: Impact of Tyres on Diffuser Performance

	No Tyres	Front Tyre	Rear Tyre	Both Tyres
Downforce Coefficient	0.426	0.254	0.035	0.039

This analysis shows that it is not enough to simply design a diffuser without the impact of the tyres being considered at any point before full car simulations are run and, as a result, counter measures were considered and implemented. Note also that the involvement of the tyres caused steady solutions in Star CCM+, the CFD package used, to experience oscillations due to the inherently unsteady dynamics of the tyre wakes not being captured, meaning that there is a variation in downforce with time and an uncertainty in its final value. Almost all of the benefit produced from the diffuser is lost from the effect of the tyres before any effects to mitigate the problem have been introduced.

2.4.3 Optimisation with Tyres

The first of the counter measures tested involved a narrower diffuser, decreased in width from 986 mm to 826 mm. The previous width was derived from the desire to have a maximum width diffuser, which was initially expected to provide maximum downforce before the interactions with the tyres were discovered, while maintaining a gap of 10 mm from the outer edge of the diffuser to the inside of the rear tyre. The reduction by 160 mm width was tested as this ensured that the widths of the front of the floor (at the front axle) and the diffuser were the same. Note that a narrow diffuser design was also used by TU Munich in the 2018 Formula Student competition [16], with this team finishing 4th overall, winning all of the dynamic events, save for the Endurance competition [17]. This development provided a performance benefit due to mitigating against the aforementioned negative effects of the rear tyre, with the downforce coefficient increasing from -0.035 to -0.092, a significant benefit. This design then became the new baseline.

Following on from this, it was decided to test a design used by competitor teams, involving the use of an upwashing device ahead of the rear wheel, which takes the form of a side diffuser, the final design of which can be seen in Figure 14. This was initially made 280 mm wide, spanning from the edge of the floor to the full width of the car, and a very significant downforce benefit was achieved, with the downforce coefficient increasing from -0.092 to -0.325. These upwashing devices produce a high static pressure on their pressure surface, which is enhanced by the blockage created from the rear tyre just behind them which spills on to the pressure surface of the side diffusers. The underside of this diffuser separates, as one would expect given that the underside flow, fed straight from the front wheel, is particularly low energy. Given this was expected beforehand, it was decided to use a high angle of expansion to gain more performance from the top surface via higher static pressure. A 20 degree angle of expansion was used compared to the 13 degree expansion for the main diffuser. Due to the high static pressure on the top surface, there is a vortex produced along the outboard edge of the device, which adds upwash to the flow on the underside of the side diffuser and ensures there is some expansion, creating a lower static pressure on the suction surface.

Due to the strong performance of the side diffusers, it was decided to increase their width further inboard, right up to the side of the chassis, while correspondingly narrowing the entire main floor and diffuser to the maximum width of the monocoque. This provided a further increase in downforce via the same method as that outlined above, with the downforce coefficient increasing from -0.325 to -0.536. Given the time constraints, there was not ample opportunity to optimise this design despite the obvious benefits it offers - for instance, a variety of different angles could have been tested and the optimal one found, perhaps yielding an even greater downforce benefit.

At this point, the reader may question why attempts were not made to prevent the wheel wake from entering the gap between the side diffusers and the ground to allow the full expansion to be realised. However, due to the fact that the regulations do not allow side skirts, for instance, it is not possible to seal the floor-diffuser underside surfaces from the tyre turbulence. What is required is a vortex from the front wing that is powerful enough to push the lower front tyre wake outboard, such as that used in Formula One [18]. However, given the low car velocities, the vortices are weak, while analysis is required under a range of cornering conditions to ensure that the flow under the floor is always devoid of the wheel wake. This requires a variety of full car simulations to be conducted, which are time-consuming. In this sense, designing for sub-optimal flow on the suction surface of a device is a conservative, yet more fail-safe design. Note that further methods to control the impact of the tyres, such as the use of side flow deflectors, are discussed by Loynes [11].

2.4.4 Relaxation of Constraints

As the full car CAD model was being assembled and various component designs were finalised, it became clear that three would be ample space to expand the diffuser from further upstream than previously thought. This arose from the fact that the lowermost spaceframe member seen in Figure 5 ended up being placed ahead of the sprocket, rather than behind it, while the sprocket itself was slightly smaller than expected due to the requirement for an exact number of teeth to be used. Further to this, the use of a rounded expansion for the diffuser means that the initial expansion is very gentle with a slow height increase over a reasonable length. With all of these factors in mind, it was decided to move the diffuser expansion forward to 470 mm ahead of its trailing edge, an increase of 180 mm. Given the desire to round the diffuser to its trailing edge, this resulted in a larger radius for the expansion, making the curvature more gentle and lowering the probability of flow separation. This can be seen in Figure 12 where, for the greater expansion length, the wall shear stress remains positive along the entire span of the diffuser (half diffuser shown), whereas there is a small amount of reversed flow in the centre for the shorter expansion, as expected given the 3D design criteria mentioned in Section 2.4.1. Overall, this suggests that further optimisation could be achieved by increasing the diffuser expansion angle on the final design, gaining further downforce.

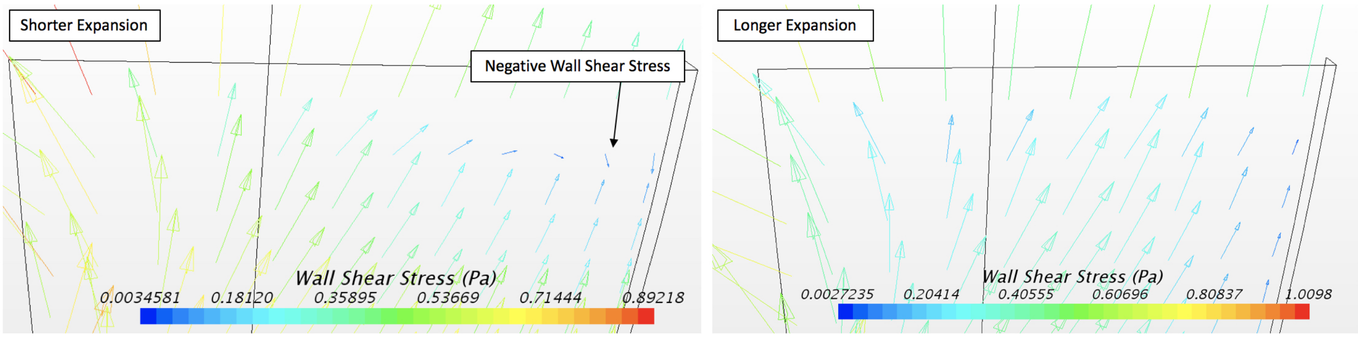

Figure 12: Wall shear stress on diffuser underside for two different expansion lengths

2.4.5 Extreme Conditions

During the main part of the design process, the diffuser was always tested at a 15 m/s flow velocity, assuming that the ground clearance would remain very similar to that at static, 30 mm. However, as the speed increases, the diffuser was expected to experience stressed conditions, whereby its performance is hindered by the attitude of the car in terms of its roll, pitch (positive nose-up) and generally lower ground clearance. To account for this, the final diffuser design was tested under three additional conditions; at maximum speed (45 m/s) just before the brakes are applied (Case 1) and just after the brakes have been applied (Case 2), as well as at the maximum corner apex speed of 40 m/s (Case 3). Using the suspension stiffness and lap simulator ride height plots, Jones [9] produced the necessary car parameters for these CFD simulations to be run, and they are given in Table 3, together with the resulting downforce coefficients.

Table 3: Conditions for stress-testing the diffuser

	Front Ride Height /mm	Rear Ride Height /mm	Pitch Angle /deg	Downforce Coefficient
Static	30	30	0.00	0.570
Case 1	24	12	0.44	0.840
Case 2	13	24	−0.41	0.600
Case 3	21	21	0.00	0.576

The end of straight test, under braking, results in the car being particularly close to the ground at the front of the floor, thereby reducing the mass flow under the floor, while also putting the entire floor at an angle of attack, so there are effectively two diffusers. Furthermore, the maximum speed test before braking causes the car to pitch nose up and reduces the ground clearance at the beginning of the diffuser expansion, which should increase the severity of the suction peak and potentially cause an increased adverse pressure gradient as the diffuser attempts to expand the flow back to ambient pressure by the time it reaches the trailing edge. Meanwhile, it is necessary the diffuser does not lose significant performance in the highest speed corners due to the instability that would result, in the form of oversteer. Surprisingly, none of the three conditions produced a decrease in the downforce coefficient. This is perhaps due to the fact that, as mentioned above, the final diffuser design is slightly conservative, so it can handle more aggressive orientations without separating. Of interest is the significant downforce gain achieved by having the rear of the car lower than the front - with this concept, the entire floor is acting like a converging duct, accelerating the flow underneath and reducing its static pressure - it would be interesting to explore this design concept further in future iterations of the design.

2.5 Manufacturing, Mass Estimation and Regulatory Details

The diffuser will be manufactured out of a carbon fibre reinforced poylmer, with a quasi-isotropic lay-up. Lee [19] has more information on this. Finite Element Analysis of the diffuser was attempted in Creo Simulate using the standard carbon fibre-epoxy composite material within this software, which Professor Greenhalgh [20] confirmed had a reasonable Young's Modulus for a quasi-isotropic lay-up. However, despite using a number of mounting points between the monocoque and diffuser (mainly to the underside, but also to the side diffusers - see Figure 14) and varying the mesh size in Creo, the mass of the diffuser required to pass the two deflection tests in the rules (a 50 N point load must not result in a deflection greater than 25 mm and a 200 N distributed load over a 225 cm^2 area must not cause a deflection greater than 10 mm [21]) was very high, at over 20 kg, with a material thickness of almost 10 mm.

On the contrary, the 2018 Group Design project produced a diffuser with a mass of just 5 kg, and a total composite lay-up thickness of 2 mm. Although this diffuser is larger, and therefore should have a greater mass, a surface area comparison between the two designs suggested that this diffuser should weigh approximately 10 kg, and hence this value will be considered in the overall performance estimation of this aerodynamic device, while it was also used for the material cost. Moreover, the regulations have stipulations regarding the rounding of the edges of any part of the car which can contact a pedestrian, with rounds of radius 3 mm and 5 mm for vertical and horizontal edges respectively that are forward facing and 1 mm for non-forward facing edges [21]; this diffuser design incorporates these regulatory constraints.

2.6 Full Car Performance

Despite the fact that the aerodynamic devices were developed in their own right, it is ultimately the interaction between them that determines how they perform on a full car model. With this in mind, the downforce coefficient from the floor and diffuser when tested with other components was 0.377, 34% lower than that achieved from testing the floor and diffuser with tyres only (0.570). A significant difference between the flow with and without other devices was the wake off the rear of the car, behind the engine, which produced a low static pressure region above the diffuser, significantly reducing downforce. Further details on the full car simulation are given by Singh [14].

Overall, the points gain in the Formula Student dynamic events is 11 points, with the following breakdown: a 21.7 point improvement from a downforce coefficient of 0.377, a 0.7 point loss from a drag coefficient of 0.027,

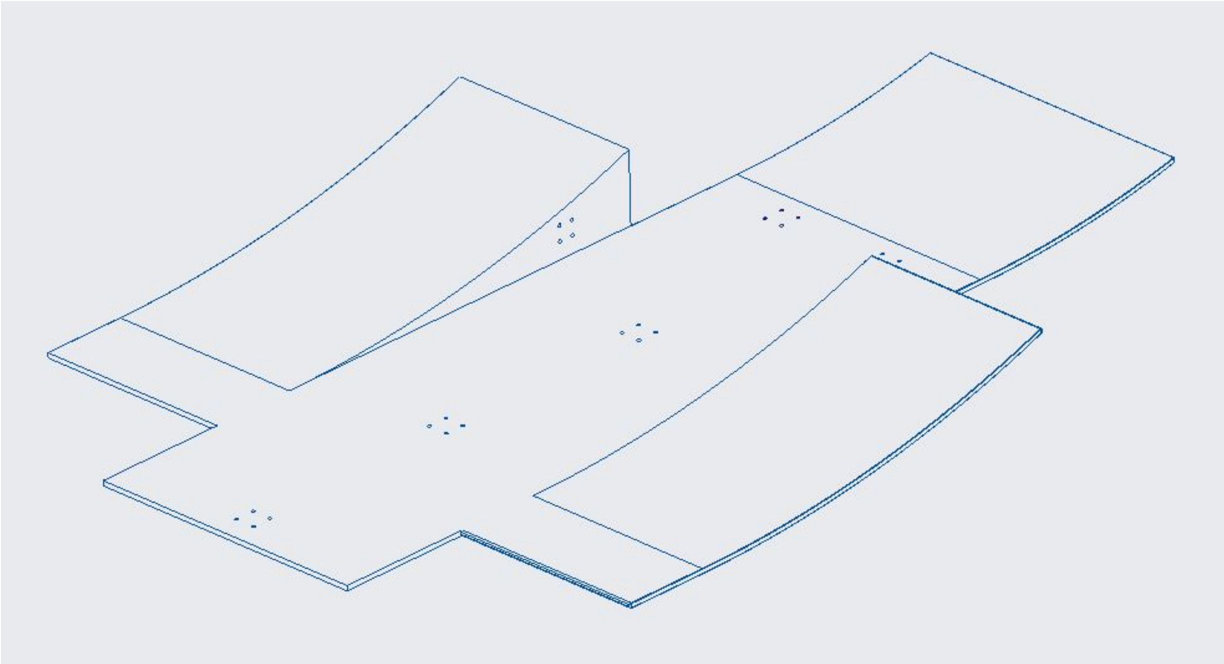

Figure 13: Final Diffuser Design

and a 10 point loss from a mass of 10 kg. The cost of the diffuser, including material and manufacturing comes to £900, a breakdown of which can be found in [19].

2.7 Future Work

As has been mentioned, a significant amount of the design work was carried out while giving maximum freedom to other parts of the car, and hence, little optimisation was done on the final configuration for the main diffuser, resulting in a conservative design, that has more performance potential. Furthermore, due to the later addition of the side diffusers, they too could benefit from detailed optimisation. Most testing was done using a straight on velocity condition, with zero pitch angle - for a more optimised design, a greater range of conditions should be used when testing CFD. However, a key limitation of CFD, compared to the wind tunnel is that the time taken to run simulations is much longer. With track running, it would be possible to verify that the results from the lap simulator in terms of ride height and pitch angle and subsequently, development should be conducted at each of the key conditions before a part is assessed to be positive. Given that a baseline design has already been achieved, there are fewer degrees of freedom available, permitting such a test plan to come to fruition. Finally, despite the narrowing of the diffuser, the rear tyre wake is interacting with the outboard underside of the diffuser, and preventing this could yield further benefit. Loynes [11] tested a vertical fence on the edge of an earlier diffuser design, but the benefit was not judged to be significant enough to account for the extra cost and mass. However, given that other teams use such a design, as well as further fences inboard, this could form an interesting development path in future.

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Appendices

A Diffuser Three View Technical Drawing

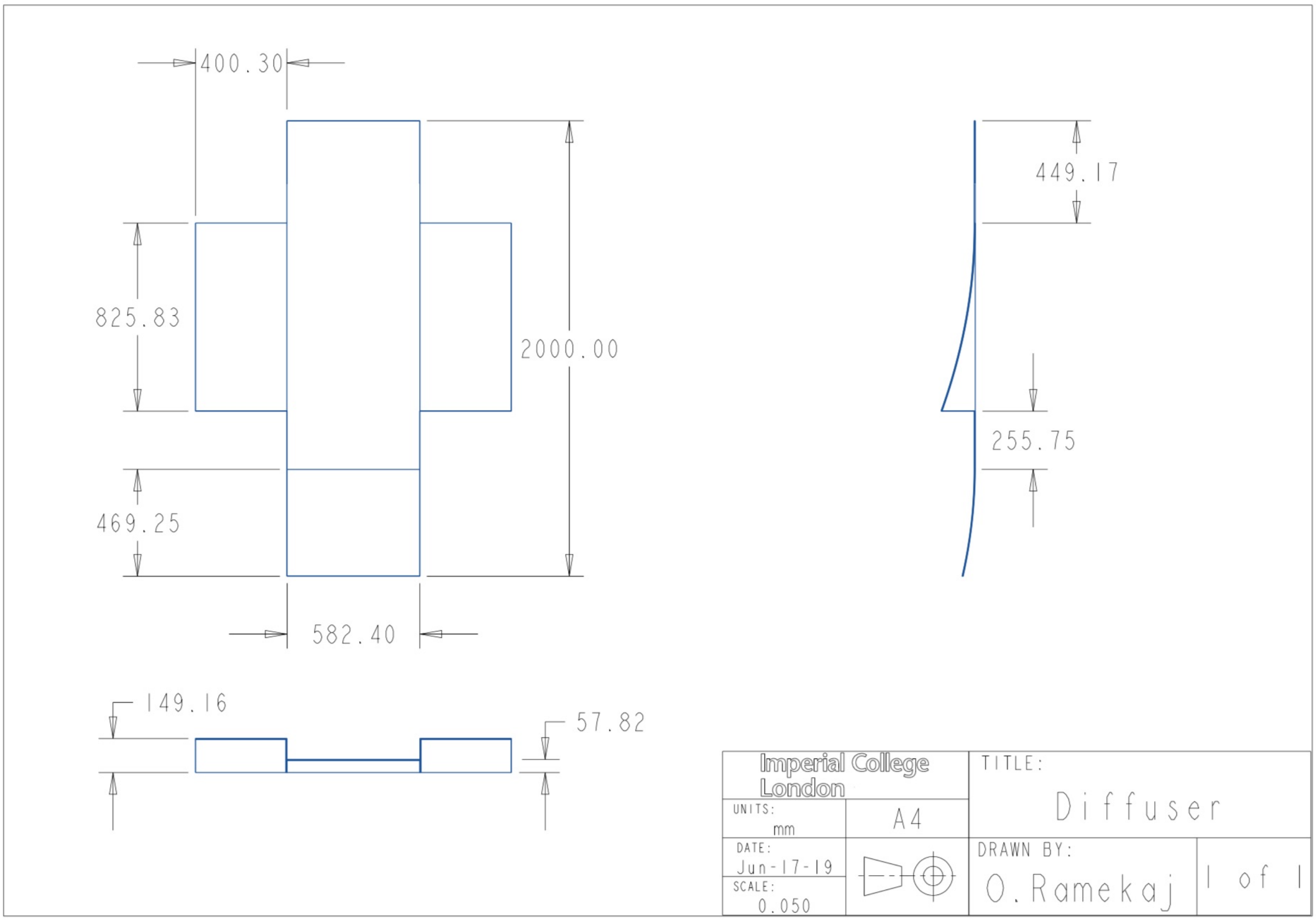

Figure 14: Final Diffuser Three View Drawing by Ramekaj [1]